

Fig. 2: An example with defined circuit parameters (Courtesy: www.ni.com)

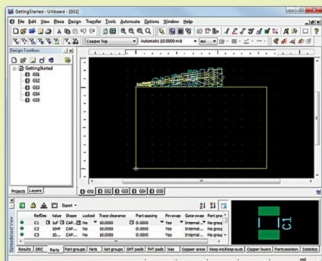


Fig. 3: User interface of NI Ultiboard

sweep and Monte Carlo analysis to validate design corner cases. The user can also sweep critical component parameters and iterate through parametric information in advanced simulations to optimise design performance.

Simulation enhancement with over 26,000 devices. Analogue designers rely heavily on datasheets provided by component manufacturers to design a circuit to specification. Although datasheets include critical information of individual components (and remain an indispensable resource), they lack information of how parts will behave within different design configurations. Devices from Maxim, Infineon, NXP, ON Semiconductor and Analog Devices have been added to the library of the latest SPICE models to evaluate design choices.

Rapid prototyping with user-definable templates. User-defined templates for circuit schematics and PCB layouts provide a comprehensive starting point to engineers to start designing their circuits and printed circuit boards. With a template, various elements such as connector placement, critical devices and board layout can be pre-configured to aid in the design process. These templates can save significant amounts of time spent on designs by collaborating with others, providing set constraints for adjacent

Power circuits' simulation with IGBT models and EPC devices. The powerful Multisim simulation enables the evaluation of power circuits of different ratings at an early design stage. To accurately model power electronics systems, you can use models for IGBT and MOSFET switches, different active and passive components, and switching controllers. With the addition of new eGaN MOSFETs from market innovator Efficient Power Conversion Corporation (EPC), and a configurable thermal IGBT model, simulation of applications with greater accuracy can be achieved.

electronics or prototyping for hardware from National Instruments. You can also create, share and modify user-configurable templates to set critical spacing and layout settings to aid in prototyping.

Features of Ultiboard

Ultiboard, which can be easily integrated with Multisim, has some features which are special to it.

Easy-to-use design environment. The NI Ultiboard interface enables efficient layout and routing of PCB designs. Tools such as spreadsheet view, toolbox and design wizards easily manage, control and define any board layout. Part placement and copper routes are optimised to allow full control for precise definition of critical parts or automation for quick design completion. The 3D preview renders a completed, populated board to gain a virtual perspective before production. Ultiboard exports and produces industry-standard formats, such as Gerber and DXF, to take a final, optimised board to prototype and manufacture.

Layout and routing. Ultiboard features intuitive tools for manual and automated part placement and routing. The flexible push-and-shove environment combined with placement processes ensures quick layout. Trace placement in Ultiboard offers flexibility to quickly and efficiently adjust, based on the degree of design complexity.

Integrated design. Seamless integration with NI Multisim allows easy transfer of schematics to layout, to quickly define a board for manufacture. Cross-probing ensures any land pattern can be selected at board level to simultaneously highlight the Multisim schematic counterpart. Forward and back annotation ensures that changes made at either the schematic or layout level are always reflected in

Post installation

If you go through the documentation provided with the software, it will take you through all the stages of circuit design flow, from schematic capture through simulation and analysis. After following the steps outlined in the documentation, you will have designed a circuit that samples a small analogue signal, amplifies it and then counts the cycles on a simple digital counter.